



Internship report

Using AI for Emotional Analysis Based on Physiological
Data from Wearable Sensors

Internship Duration: 2025/04/21 ~2025/08/20

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Acknowledgements

I would like to express my deep gratitude to all the people who accompanied and supported me during my internship, an extremely enriching experience both professionally and personally.

First of all, I would like to thank **the IRCN** (International Research Center for Neurointelligence) for having welcomed me to its premises, allowing me to evolve in a high-level, stimulating and innovative research environment.

I would also like to express my sincere gratitude to **Ms. Yukie Nagai**, Director of the Laboratory, for granting me the opportunity to join her team. His openness, scientific vision and commitment to the field of computational neuroscience and artificial intelligence greatly enriched my experience.

My thanks then go to **Mr. Herbuela Von Ralph Dane**, my internship supervisor, for his attentive supervision, his availability and his wise advice throughout this period. His technical expertise and pedagogy have allowed me to progress and better understand the issues related to the recovery and analysis of physiological data via connected devices.

Finally, I would like to thank all the members of the laboratory for their welcome, their support and the enriching exchanges that have marked my career. Their collaborative spirit greatly contributed to making this internship a formative and unforgettable experience.

Thank you all for your trust, patience and support.

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1. Company Introduction

The IRCN (International Research Center for Neurointelligence) is a research institute of excellence located at the University of Tokyo, Japan. Founded in 2017 as part of Japan's World Premier International Research Center Initiative (WPI), IRCN aims to become a global leader in the emerging field of neurointelligence.

The IRCN explores how the brain develops intelligence and how this knowledge can transform approaches to mental health, learning, artificial intelligence and human-computer interaction.

The IRCN's approach is interdisciplinary, at the crossroads of:

- 1- Developmental neuroscience
- 2- Psychiatry and neurodevelopment
- 3- Artificial Intelligence and Machine Learning
- 4- Mathematical Modeling
- 5- Advanced cognitive technologies

A central area is the exploration of neurointelligence, a concept that links the understanding of brain mechanisms to the design of new forms of artificial intelligence. At IRCN, researchers are developing computer models inspired by the functioning of the brain, particularly the prefrontal cortex, to create AIs capable of flexible learning, reasoning and adaptation. These bio-inspired systems aim to go beyond the limits of current algorithms, by integrating properties such as working memory, contextual decision-making or neuronal plasticity.

IRCN also conducts advanced research in computational psychiatry, an emerging discipline that uses mathematical and computational tools to model the symptoms and mechanisms of mental illness. The objective is to develop objective methods for diagnosing and monitoring psychiatric disorders, based on biological, behavioral and cognitive data.

In parallel with its scientific activities, IRCN pays particular attention to the management of Social and Environmental Responsibility (CSR). The organization is implementing several concrete initiatives to reduce its environmental impact: waste is systematically sorted, a rational use of air conditioning is encouraged, and teleworking is allowed in order to limit non-essential travel. These practices reflect IRCN's commitment to integrating sustainability and well-being into its day-to-day operations, in line with social and ecological issues.

2. Introduction

I had the opportunity to join the IRCN (International Research Center for Neurointelligence), a research laboratory based at the University of Tokyo, specializing in computational neuroscience and artificial intelligence. Working in this type of interdisciplinary environment was particularly appealing to me, as I have long been interested in the interactions between the brain, human behaviour and smart technologies.

My internship focused on a topic that I was immediately passionate about: the extraction and analysis of physiological data from wearable devices, such as smartwatches, with the aim of detecting emotional states in individuals. These data include heart rate, ECG, EEG, HRV, respiratory rate, etc. — indicators often linked to unconscious emotional reactions.

To interpret these complex signals, I worked with deep learning techniques, and more specifically on **cross-modality distillation**, an approach to transfer knowledge from a powerful model based on rich modalities (such as multimodal signals) to a lighter model, capable of operating in real time on embedded devices. The final goal of this work is the development of a mobile application capable of running this AI model locally to analyze physiological data and provide an estimate of the user's emotional state. This method is particularly relevant for concrete applications such as emotional assistance, well-being or human-computer interaction.

The working environment offered by the IRCN, both academic and open to innovation, allowed me to explore cutting-edge issues while developing solid skills in signal processing, machine learning and physiological analysis. Joining an international team, oriented towards ambitious research objectives, was also a very formative human and professional experience.

3. Description of the timetable

	Lundi	Mardi	Mercredi	Jeudi	Vendredi
08:00					
09:00					
10:00	Projet 10:00-12:00	Projet 10:00-12:00	Projet 10:00-11:00	Projet 10:00-12:00	Projet 10:00-11:00
11:00			Appel Anubahv Appel Herbuela 11:00-12:00		Appel Herbuela 11:00-12:00
12:00					
13:00					
14:00	Projet 13:30-19:00	Projet 13:30-19:00	Projet 13:30-19:00	Projet 13:30-19:00	Projet 13:30-15:30
15:00					
16:00					Presentation 15:30-17:30
17:00					
18:00					Projet 17:30-19:00
19:00					
20:00					

My internship took place according to a stable and structured schedule, with a daily presence in the laboratory from Monday to Friday, from 10 a.m. to 7 p.m. During these hours, I devote myself exclusively to my project, independently or in collaboration with my tutor and other members of the laboratory.

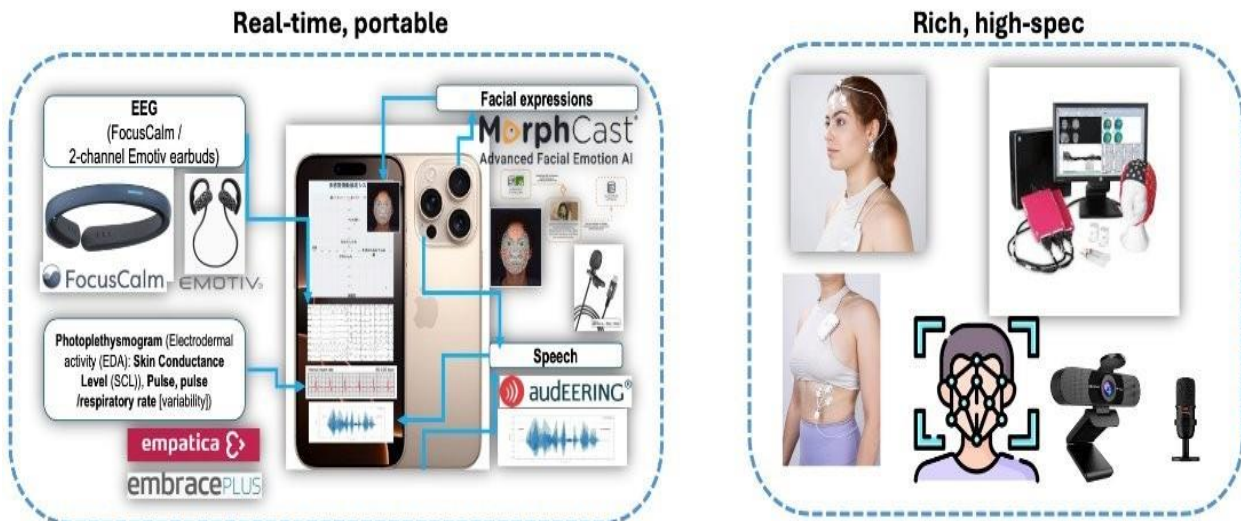
Most of my time is dedicated to the development of the emotional recognition project from physiological data, in particular signal processing, experimentation with deep learning models, and optimization in the context of a future mobile application.

In addition to this daily work, three fixed slots punctuate my week:

- 1- Wednesday from 11am to 12pm: weekly call with my internship tutor and an Indian researcher collaborating on the project. These exchanges provide an opportunity to take stock of progress and to receive research leads and scientific advice.
- 2- Friday from 11 a.m. to 12 p.m.: meeting with my tutor and all the interns in the laboratory. Each trainee presents his or her progress in the form of an oral (or visual) presentation, which allows for collective follow-up and promotes the exchange of ideas.
- 3- Friday from 3:30 pm to 5:30 pm: weekly two-part meeting with Mrs. Yukie Nagai, director of the laboratory:
 - A first time of "management meeting", where she shares the lab's news (current projects, conferences to follow, deadlines, etc.).
 - Then an in-depth presentation by a member of the laboratory on his or her research project, followed by a discussion with the director, who proposes avenues for improvement or orientation.

These regular moments are very enriching, because they allow me not only to develop my own work, but also to immerse myself in the scientific and collaborative dynamics of the laboratory.

4. Description of the missions carried out



Development and Validation of a Real-Time Multimodal Emotion Estimation System Using Wearable Sensors and Smartphones

4.1. The project

This image illustrates the project to develop a real-time emotion detection system that I was assigned to, based on wearable physiological sensors and AI algorithms embedded on smartphones.

The idea is to exploit several sources of biological and behavioral signals (EEG, facial expression, voice, heart rate, breathing, etc.) to estimate an individual's emotional state in a detailed and personalized way. The challenge is to make this solution work in real time on a mobile phone, despite resource constraints.

Multimodal data sources used:

1. EEG (electroencephalogram):

- Measured with lightweight sensors like FocusCalm or Emotiv headphones (2 channels).
- Provides an estimate of brain activity related to concentration or stress.

2. Facial expressions:

- Video analysis via MorphCast AI to detect emotions from facial movements.

3. Voice (intonation and rhythm):

- Real-time acoustic analysis with audeERING to identify voice-expressed emotions.

4. Peripheral physiological signals:

- Captured with smartwatches like EmbracePlus (Empatica) and Apple Watch.
- Data includes:
 - Heart Rate
 - Heart Rate Variability (HRV)
 - Respiratory Rate
 - Electrodermal Response (EDA/SCL)

4.2. My projects (quick explanation)

1. Smartwatch Benchmarking:

First, I carried out a comparative analysis of two connected watches: the Empatica EmbracePlus and the Apple Watch, in order to determine which was the most suitable for the acquisition of physiological data for our use case. This evaluation was based on several criteria: types of sensors available, signal quality, sampling rate, access to real-time data, and software compatibility.

The Apple Watch has proven to be easier to integrate into an iOS environment for real-time retrieval of data such as heart rate, heart rate variability (HRV), and respiratory rate.

2. Development of an iOS application for the collection of physiological data

Following this evaluation, I developed a native iOS application that allows you to connect the Apple Watch to an iPhone and retrieve the physiological data mentioned in real time.

The app uses HealthKit and WatchConnectivity to establish communication and ensure smooth synchronization between the watch and phone.

The data is then formatted and stored to feed into emotional AI models, with the goal of enabling inference directly from the smartphone.

3. Knowledge Distillation and Cross-Modality Distillation

Once physiological data was retrieved using a smartwatch (in my case, heart rate and heart rate variability), the goal was to design an artificial intelligence model that could detect emotions, while still being lightweight enough to work in real-time on low-resource devices, such as a smartphone.

I have set up a knowledge distillation approach, extended to a form of Cross-Modality Distillation. Several AI models were trained with different teachers. Two databases were used: YAAD, which is based on ECG signals, and DEAP, which is based on PPG signals. Both provide annotations in valence and arousal, allowing the determination of the emotional state.

4. AI personalization via user data

Finally, I experimented with a form of model customization based on each user's data.

By integrating individual feedback (self-assessments, feelings, etc.) or simply by taking into account the intra-individual variability of cardiac signals, I adjusted the models so that they were more sensitive to the physiological specificities of each person.

This improves the robustness and accuracy of the system, allowing for continuous adaptation to the real user.

5. Results of the missions carried out

5.1. Comparative evaluation of connected watches:

Initially, I began my experiments with the Empatica EmbracePlus watch, a medical device designed for research, capable of measuring several physiological signals such as EDA (electrodermal activity), temperature, acceleration, and the PPG signal to extract heart rate.



However, during testing, the watch generates two types of data files: CSV files and Avro files. However, despite my efforts to recover and analyze the data, including with the help of Empatica's technical team, the resulting CSV files proved to be unusable: `device_not_recording`

Here are the different sensors of each file:

CSV file

Steps

Accelerometer Standard Deviation

Avro Spins

Steps

Accelerometer

CSV file	Avro Spins
Actigraphy Counts	/
Activity Counts	/
Body Position	/
Metabolic Equivalent of Task	/
Pulse Rate	/
Skin Temperature	Temperature
Acticounts	/
Activity Classification	/
Activity Intensity	/
Electrodermal Activity (EDA)	ElectroDermal Activity (EDA)
GRP	/
Respiratory Rate	/
Sleep	/
Wearing Time	/
/	Gyroscope
/	Blood Volume Pulse (BVP)
/	Tags
/	Systolic Peaks

We used Empatica's APIs as well as the AWS Boto3 library to access Empatica data stored on the AWS Cloud. Using these tools, I developed code to retrieve, view, and modify data files directly from storage. This makes it much easier to manage and analyze biometric data.

Photos of the different types of files saved in the empatica cloud:

```
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_activity-classification.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_activity-counts.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_activity-intensity.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_body-position.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_eda.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_met.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_prv.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_pulse-rate.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_respiratory-rate.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_sleep-detection.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_step-counts.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_temperature.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/digital_biomarkers/aggregated_per_minute/1-1-0004_2025-05-01_wearing-detection.csv
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/raw_data/v6/1-1-0004_1746073513.avro
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/raw_data/v6/1-1-0004_1746077126.avro
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/raw_data/v6/1-1-0004_1746078698.avro
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/raw_data/v6/1-1-0004_1746079089.avro
v2/2000/1/1/participant_data/2025-05-01/0004-3YK911G12T/raw_data/v6/1-1-0004_1746079260.avro
```

Here's what they were in the CSV files:

```
1745798400000 2025-04-28T00:00:00Z 2000-1-1-0002 nan device_not_recording
1745798460000 2025-04-28T00:01:00Z 2000-1-1-0002 nan device_not_recording
1745798520000 2025-04-28T00:02:00Z 2000-1-1-0002 nan device_not_recording
1745798580000 2025-04-28T00:03:00Z 2000-1-1-0002 nan device_not_recording
1745798640000 2025-04-28T00:04:00Z 2000-1-1-0002 nan device_not_recording
1745798700000 2025-04-28T00:05:00Z 2000-1-1-0002 nan device_not_recording
1745798760000 2025-04-28T00:06:00Z 2000-1-1-0002 nan device_not_recording
1745798820000 2025-04-28T00:07:00Z 2000-1-1-0002 nan device_not_recording
1745798880000 2025-04-28T00:08:00Z 2000-1-1-0002 nan device_not_recording
1745798940000 2025-04-28T00:09:00Z 2000-1-1-0002 nan device_not_recording
```

So I fell back on the avro files. An Avro file is a compact binary format used to store structured data with a built-in schema. Empatica's biometric data is often exported to Avro, but this format is difficult to read directly. To be able to analyze this data easily, you need to use Empatica's APIs that allow you to convert Avro files into CSV files, which are easier to manipulate in traditional tools.

Here's an example of an avro file converted to a CSV file:

```
unix_timestamp,bvp
1746001941966968,-0.2791750431060791
1746001941982593,-0.2795398533344269
1746001941998218,-0.2777690291404724
1746001942013843,-0.27440762519836426
1746001942029468,-0.27010226249694824
1746001942045093,-0.26555317640304565
1746001942060718,-0.2614591419696808
1746001942076343,-0.2584681808948517
1746001942091968,-0.25713783502578735
1746001942107593,-0.25790640711784363
1746001942123218,-0.26107561588287354
1746001942138843,-0.2667984366416931
1746001942154468,-0.27506816387176514
1746001942170093,-0.28571492433547974
1746001942185718,-0.29841962456703186
1746001942201343,-0.3127528131008148
1746001942216968,-0.32822859287261963
1746001942232593,-0.34434252977371216
1746001942248218,-0.3605641722679138
1746001942263843,-0.3762887716293335
1746001942279468,-0.39080023765563965
1746001942295093,-0.4033060073852539
1746001942310718,-0.41304102540016174
1746001942326343,-0.4193834066390991
1746001942341968,-0.4219472110271454
1746001942357593,-0.420642614364624
```

Real-time data retrieval was not possible with the Empatica system, as CSV files were only transferred every 10 minutes and Avro files every 30 minutes. This latency did not suit our needs. So we decided to switch to the Apple Watch, which offers much faster and real-time tracking and data transfer.



Table of the different sensors of the Apple Watch 10 series:

Sensor Type	Real-time Periodic	
Heart Rate Sensor	Yes	Yes
Accelerometer	Yes	Yes
Gyroscope	Yes	Yes
Heart Rate Variability (HRV)	No	Yes
SpO2 (Oxygen)	No	Yes
Respiratory Rate	No	Yes
Skin Temperature	No	Yes
Blood Oxygen Sensor (SpO2)	No	Yes
ECG	No	Yes

5.2. Development of an iOS application for the collection of physiological data

The goal of this project is to develop an app for Apple Watch and iPhone that collects physiological signals in real time. These signals will be used to build an emotion recognition system. Artificial intelligence will analyze the data to identify the user's emotional states.

5.2.1. Collection of heart rate data

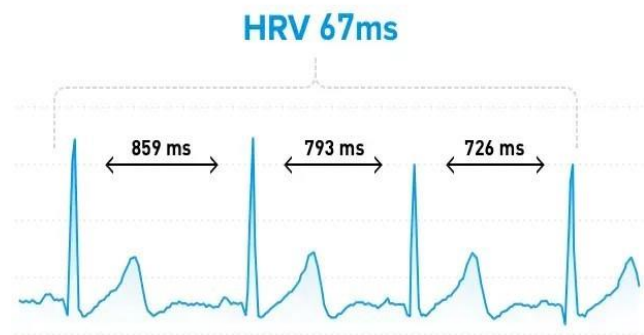
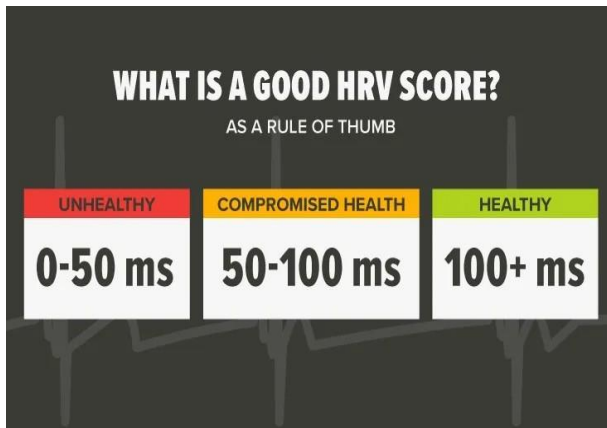
The first step in the project is to collect real-time heart rate data from the Apple Watch. Here are the normal values for different types of individuals.

Age Group	Resting Heart Rate (Beats Per Minute)	During Moderate Activity	During Vigorous Activity
Newborns (0-1 month)	70-190	-	-
Infants (1-11 months)	80-160	-	-
Children (1-2 years)	80-130	-	-
Children (3-4 years)	80-120	-	-
Children (5-6 years)	75-115	-	-
Children (7-9 years)	70-110	-	-
Children (10-12 years)	60-100	-	-
Teenagers (13-15 years)	55-95	100-160	130-190
Young Adults (16-25 years)	60-100	100-170	140-200
Adults (26-35 years)	60-100	95-162	135-195
Adults (36-45 years)	60-100	93-157	130-190
Adults (46-55 years)	60-100	90-153	125-185

5.2.2. Calculating HRV (Heart rate variability)

The second step is to calculate the HRV in real time from the heart rate data collected by the Apple Watch in order to assess emotional states.

HRV is the natural variation in time between two consecutive heartbeats.



The RR interval is the time between two consecutive heartbeats, measured in milliseconds. It reflects the natural variation in heart rate and is essential for calculating heart rate variability (HRV).

How to calculate it:

$$\text{Heart Rate (BPM)} = \frac{60\,000}{\text{RR Interval (ms)}}$$

Using RR intervals, we can calculate RMSSD (Root Mean Square of Successive Differences) or SDNN, which are common measures representing heart rate variability (HRV).

<https://www.kubios.com/blog/hrv-analysis-methods/>

$$\text{RMSSD} = \sqrt{\frac{1}{N-1} \sum_{n=1}^{N-1} (\text{RR}_{n+1} - \text{RR}_n)^2}$$

$$\text{SDNN} = \sqrt{\frac{1}{N-1} \sum_{n=1}^N (\text{RR}_n - \overline{\text{RR}})^2}$$

5.2.3. Calculating Respiratory Rate with CoreMotion

In this step, the respiratory rate is estimated by analyzing the subtle movements caused by breathing thanks to the accelerometer, which detects variations in the gravitational force on the wrist. By following these periodic movements, the app can deduce the number of breaths per minute. This method offers a non-invasive way to monitor breathing without additional sensors.

NORMAL RESPIRATORY RATE BY AGE

AGE	BREATHS PER MIN
NEWBORNS	44
INFANTS	20-40
PRESCHOOL CHILDREN	20-30
OLDER CHILDREN	16-25
ADULTS	12-20
ADULTS EXERCISING	35-45

I mainly used this github project below to make this part:

<https://github.com/shubham1chawla/HeartRespSensor>

5.2.4. Generating and Synchronizing Dated CSV Files

This Swift-based iOS module allows for the local collection and storage of structured data in CSV files, each of which is automatically named with the date of registration. Designed for simplified file management, the app allows the export and import of these files directly to the device. The data collected is intended for use in an emotion recognition AI system developed by my professor, Dr. Von, making accurate logging a must. This configuration ensures a reliable flow of data from mobile input to machine learning processing.

5.2.5. Real-time cardiac data streaming via a custom socket server

This project integrates a custom Python socket server running on Windows and a Swift-based iOS client application to enable smooth, real-time transmission of biometric data. The iOS app collects heart rate (HR) and heart rate variability (HRV) measurements from an Apple Watch every 5 seconds. These values are instantly transmitted to the desktop server over a local network connection, ensuring minimal latency and consistent transmission. The system was designed to provide immediate access to physiological signals for analysis, visualization, or AI integration.

5.2.6. Explanation

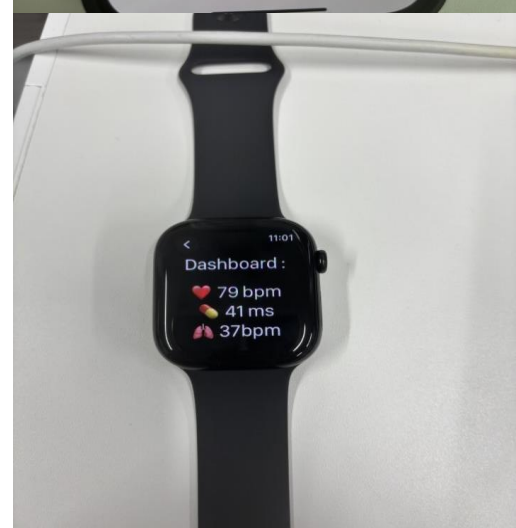
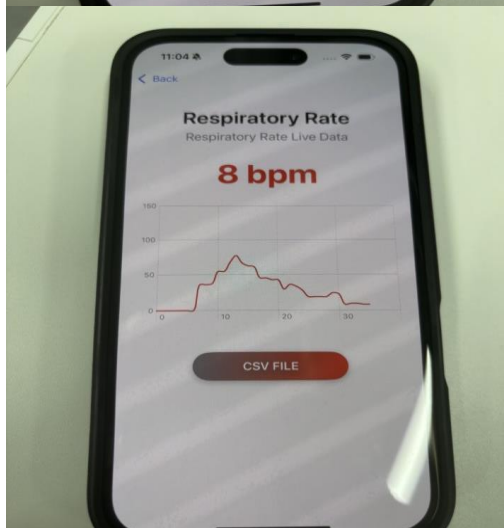
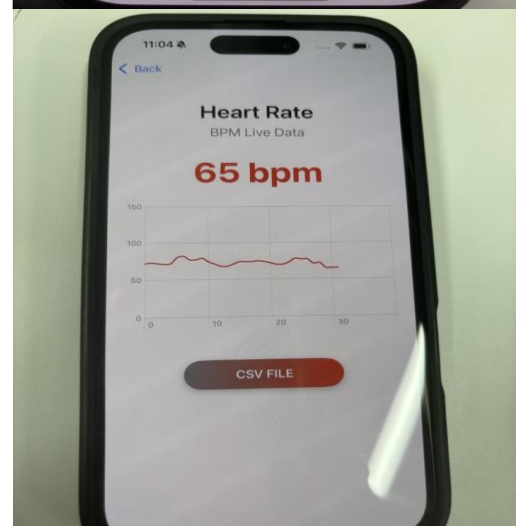
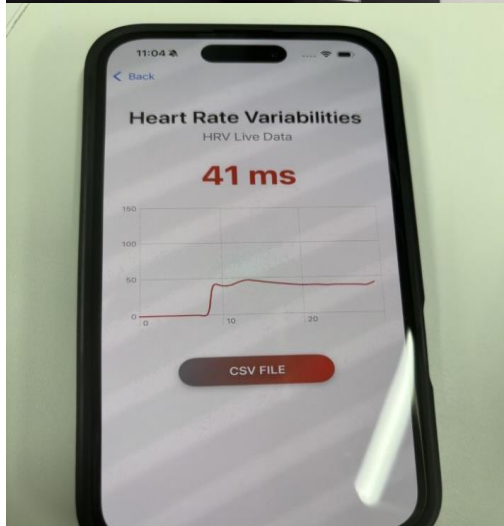
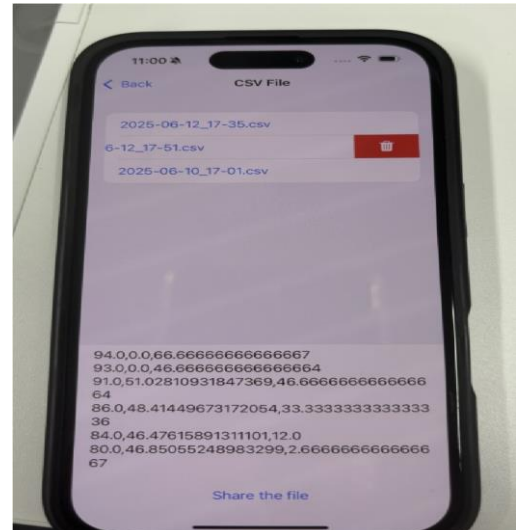
The mobile app was developed in Swift, on macOS using the Xcode development environment. This technological choice was necessary to allow a smooth integration with Apple devices, in particular the Apple Watch, used for the recovery of physiological data.

However, it's important to point out that I had no prior experience in mobile development with Swift, macOS, or Xcode. This situation represented a significant challenge at the beginning of the internship. Learning the language, development environment, and Apple APIs (such as HealthKit and CoreMotion) at the same time took a significant amount of time to get used to.

Of the five steps defined in the development of the app, step 3, which consisted of calculating the respiratory rate from the movements detected via CoreMotion, could not be finalized. Despite the effective recovery of respiratory rate values, they are extremely unstable, fluctuate sharply, and deviate considerably from the real values. They cannot therefore be used reliably in a real-world context.

As it stands, the app uses heart rate (Heart Rate) and heart rate variability (HRV) data, retrieved in real time via the Apple Watch, at a new value every five seconds. This physiological data is used as input to the embedded AI model, in order to estimate the user's emotional state.

5.2.7. Project visuals



5.3. Knowledge Distillation and Cross-Modality Distillation

5.3.1. Explanation

As part of this project, the objective is to develop an artificial intelligence model capable of detecting a user's emotional state in real time, based on physiological data retrieved from a connected watch, such as the Apple Watch. The main challenge is to design a lightweight solution, deployable on a smartphone, and not very intrusive, in order to guarantee smooth use in daily life without inconveniencing the user.

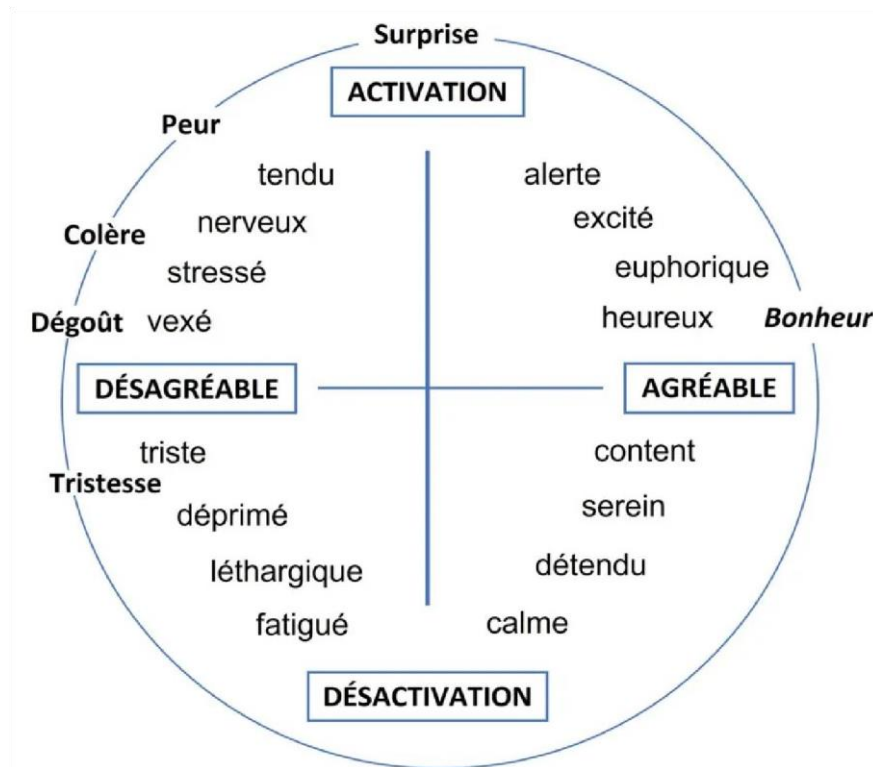
Currently, the laboratory already has a high-performance AI model, but it is based on the analysis of raw signals from many sensors (ECG, breathing, movements, etc.) and requires significant computing power, which makes it unsuitable for mobile use. In addition, wearing multiple sensors makes the system uncomfortable in the long run.

This project therefore seeks to reduce the necessary sensors to the essentials — in this case, only the heart rate extracted from an Apple Watch — while maintaining acceptable performance. The idea is to transfer intelligence from the complex model to a lighter model, capable of operating autonomously on a phone, and without depending on a bulky device.

Moreover, rather than a classification into discrete emotions (such as joy, fear, anger), this project adopts a regression approach (teacher and student). In concrete terms, this means that the model does not predict a fixed emotion, but estimates two continuous values: the valence and the arousal (Russell's model).

- Valence represents how positive or negative an emotion is (from sad to joyful, for example).
- The arousal reflects the level of physiological activation (calm to excited or stressed).

Regression here is a type of machine learning that can predict a numerical value (rather than a category), resulting in a finer and more nuanced estimate of the user's emotional state. This approach is particularly well suited to physiological signals such as heart rate, which vary continuously over time.



To comply with the computational constraints imposed by mobile devices, the project relies on a learning technique called knowledge distillation. This method consists of transferring knowledge from a complex model to a simpler model.

- 1- The “teacher” model is a large neural network, capable of analyzing physiological signals in depth and accurately predicting the emotional state.
- 2- The “student” model is a much lighter network, optimized to run in real time on a smartphone.

Being more compact, the student is less accurate when trained only with ground truths (measured or annotated valence and arousal values). To improve his performance, he also learns to imitate the teacher's predictions, which are often closer to reality than raw labels.

In concrete terms, the teacher is used only once to generate predictions on all training data. These predictions then serve as "pseudolabels" that the student will try to reproduce. The student's training is therefore based on a double supervision:

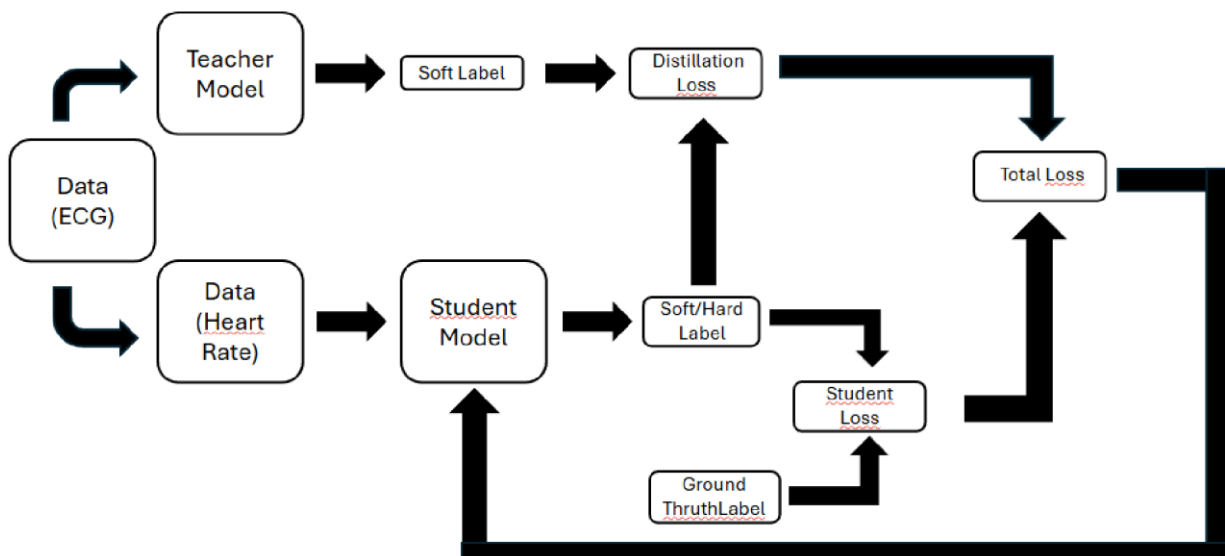
- a classic loss between his predictions and the ground truths,
- a loss of distillation between his predictions and those of the teacher.

This strategy allows the lightweight model to benefit from the richness of the complex model, while remaining fast and compact enough to work on a mobile

device. This results in a good compromise between precision, lightness and energy efficiency.

Cross-modality distillation is a particular form of this method, where the teacher and the student do not use the same types of data: for example, the teacher learns with ECG signals, while the student uses HR signals (transformed ECG) from the ECG signals. The objective is then to help the student reproduce the teacher's performance despite more limited data.

Cross Modality Distillation

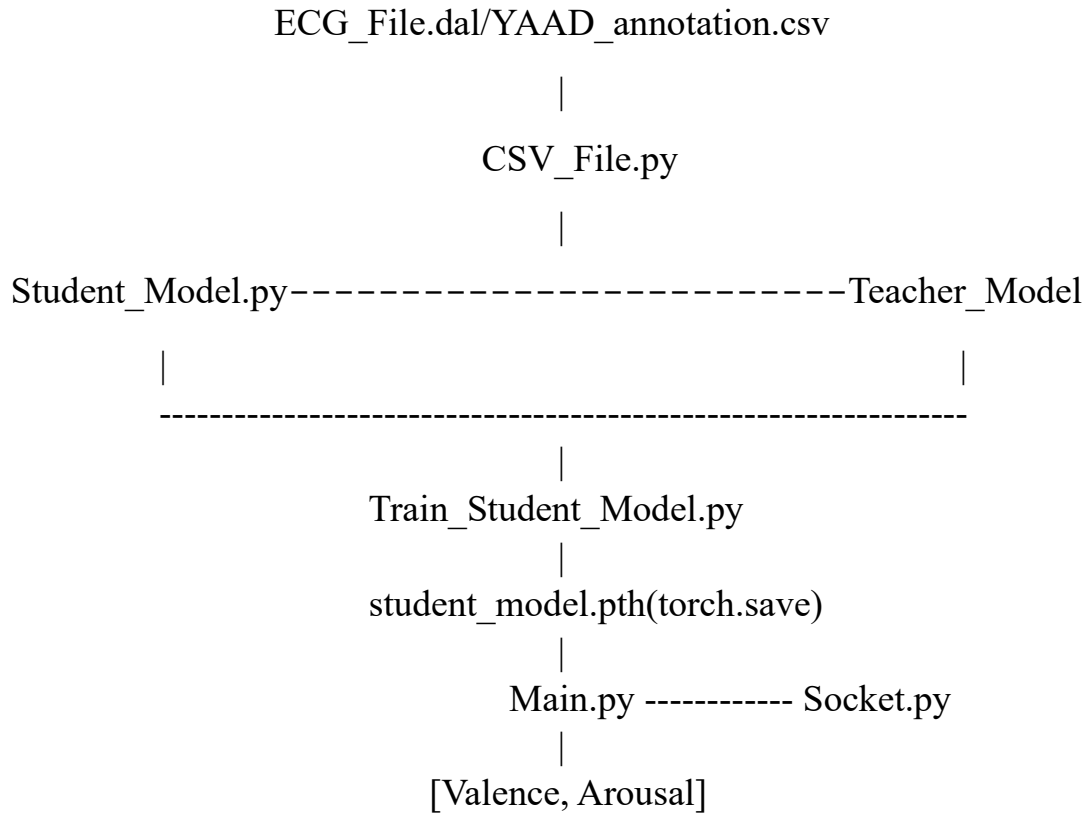


The input data is sent to both the teacher and student models.

1. The teacher model is already trained, so it is run only once to generate *soft labels*. These remain fixed during training.
2. The student model is one that is trained over several eras. At each training stage, it produces:
 - 3.1. *Soft labels* to compare with those of the teacher.
 - 3.2. *Hard labels* to be compared with ground truth.
3. We then compute two loss functions.

5.3.2. YAAD Dataset

5.3.2.1. Code Architecture

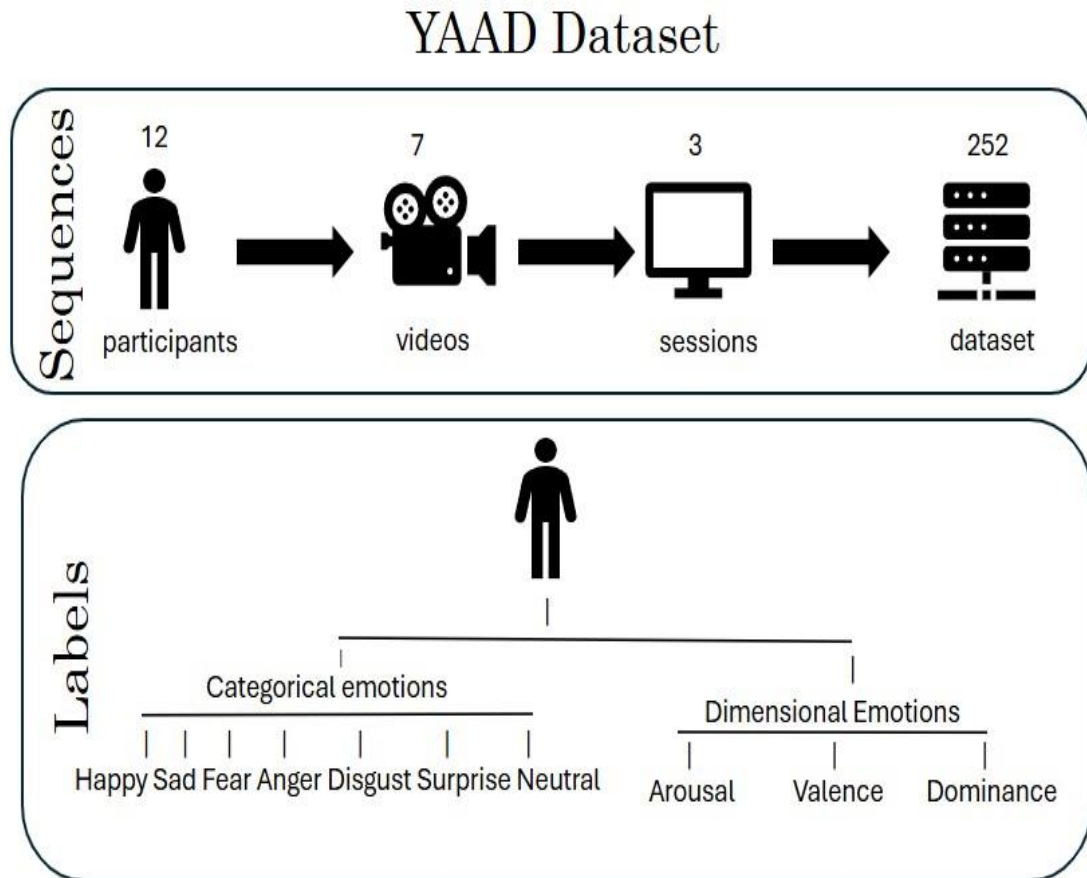


5.3.2.2. Model information

Loss function (teacher and student)	MSELoss
Adaptive Moment Estimation (optimizer)	Adam
Activation Function	Proofread
Epochs	500
Batch Size	28
Dataset Size	252
Learning Rate	1st-4th
Weight Decay	1st-5th
Shuffle	True

5.3.2.3. Results

5.3.2.3.1. Student_Model_v1



The YAAD dataset is a physiological dataset designed for the analysis of emotional states from cardiac signals. It consists of ECG (electrocardiogram) signals with a fixed length of 252 samples each. Its size and format make it a good candidate for testing lightweight architectures like a simple MLP, but also for exploring more advanced methods such as *cross-modality distillation* to see if leveraging additional signals or knowledge from a professor model can improve valence and arousal prediction.

Before introducing *cross-modality distillation*, it is important to check whether a simpler model could be enough to achieve good performance. The initial objective is therefore to set up a baseline model based on a simple MLP (*Multi-Layer Perceptron*), in order to evaluate its performance without any distillation mechanism.

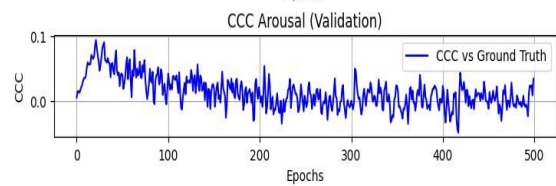
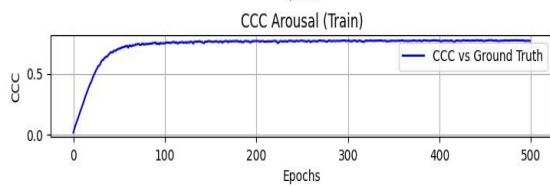
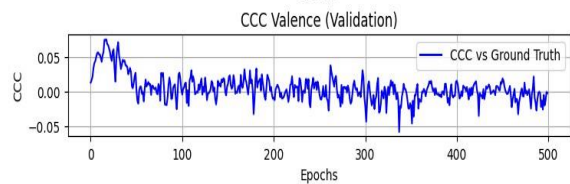
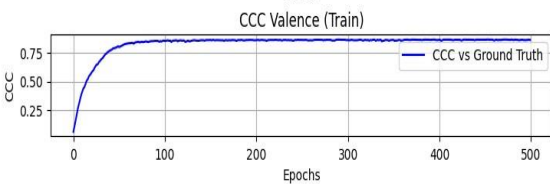
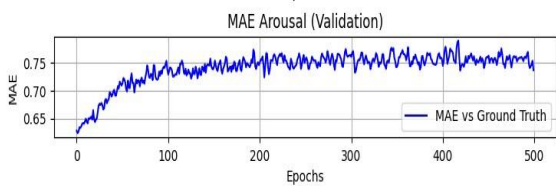
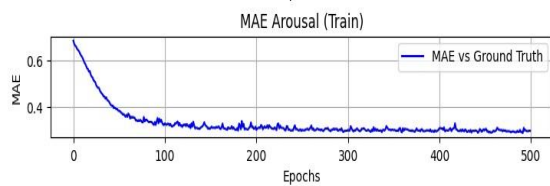
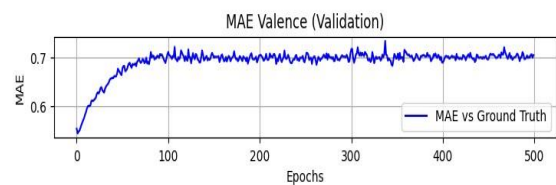
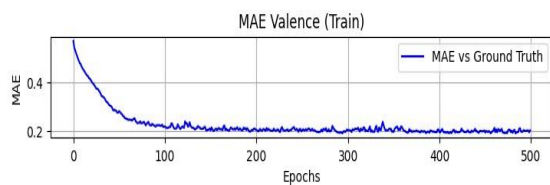
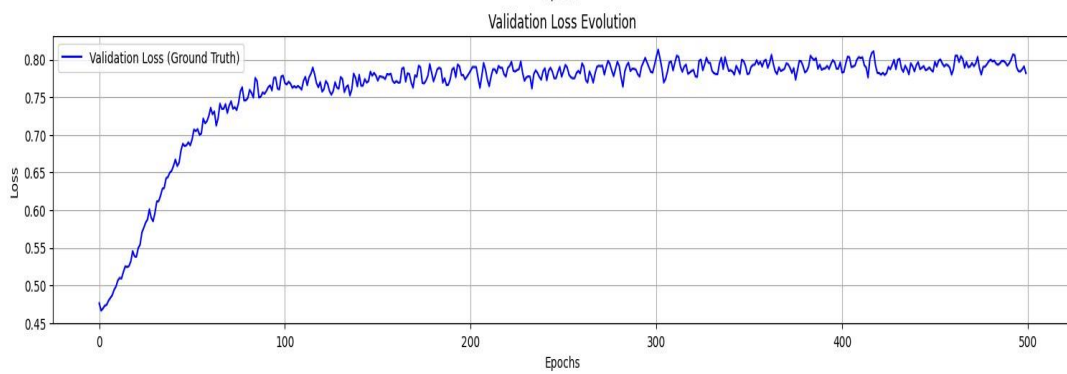
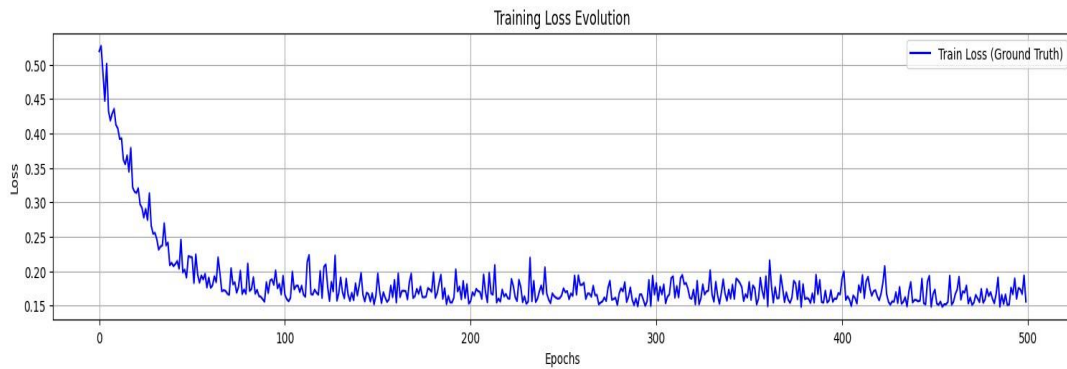
Criteria	Cross-Modality	MLP
Consumption	✗	✓
Performance	✓	⚠
Speed	⚠	✓

✓ = Clear advantage
 ⚠ = Medium / depends on context
 ✗ = Disadvantage

For this first training, the model is trained only with the student loss, i.e. the loss function associated with the target task, without using information from a teacher model. The dataset is divided into two parts: 80% for training and 20% for testing, which helps to verify the model's ability to generalize. This step aims to answer the question: *is a simple MLP already sufficient or should cross-modality distillation really be used?* By measuring the performance of the MLP, we obtain an objective point of comparison. If the MLP proves to be effective, distillation could prove superfluous; On the other hand, if the results are limited, this will justify the use of a more complex method such as cross-modality distillation. This analysis will therefore make it possible to assess the real added value of distillation compared to a simple model.

Metrics	MFA	MSE	R2_Score	CCC
Valencia (Train)	0.2037	0.1093	0.7559	0.8586
Arousal (Train)	0.3046	0.2191	0.6203	0.7603
Valencia (Val)	0.7042	/	/	-0.024
Arousal (Val)	0.7567	/	/	0.016

The results show that the model learns very well from the data with low errors (MAE, MSE) and high R^2 and CCC scores, especially for valence. However, performance drops sharply on validation data, with an MAE that increases sharply and CCC scores close to zero or even negative. This indicates overfitting: the model remembers the training data but does not generalize to the new data. It is important to look at the training graphs to visualize this difference in performance between train and validation, which clearly confirms overfitting.



5.3.2.3.2. Student_Model + Teacher

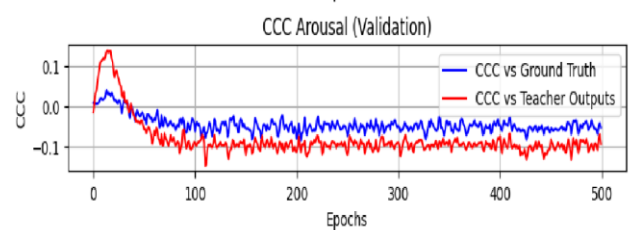
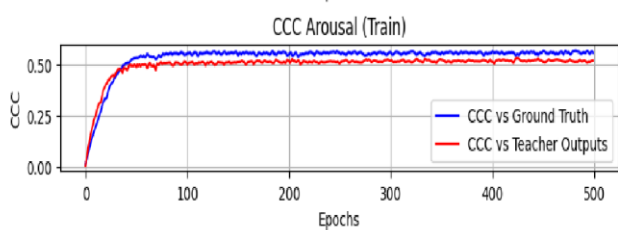
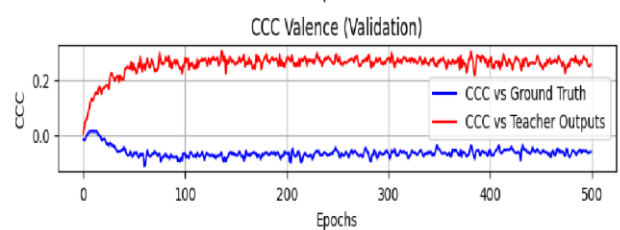
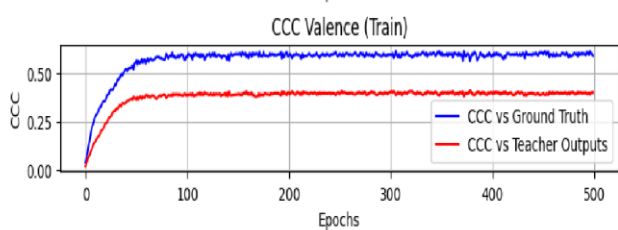
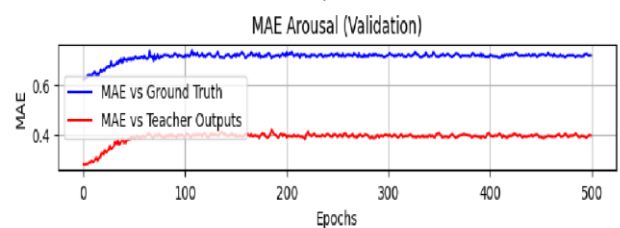
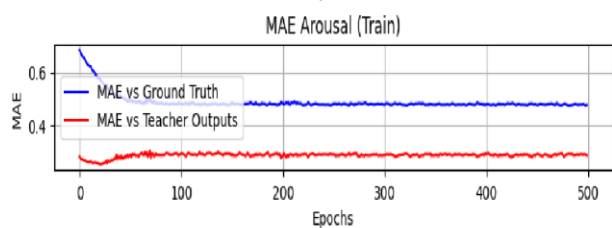
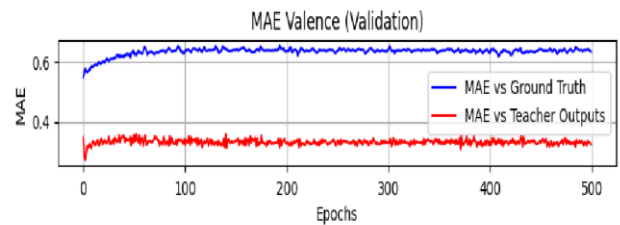
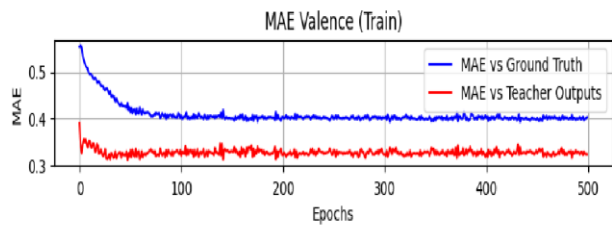
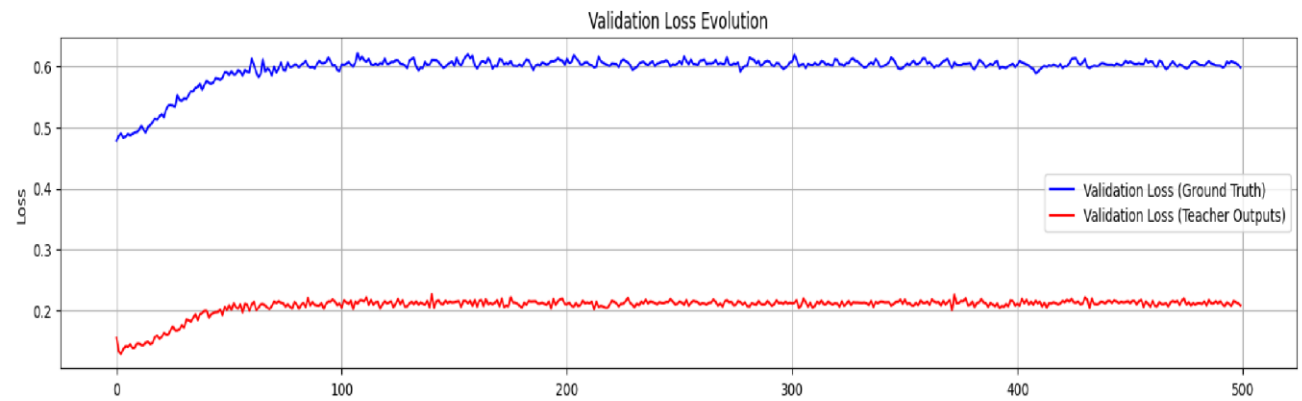
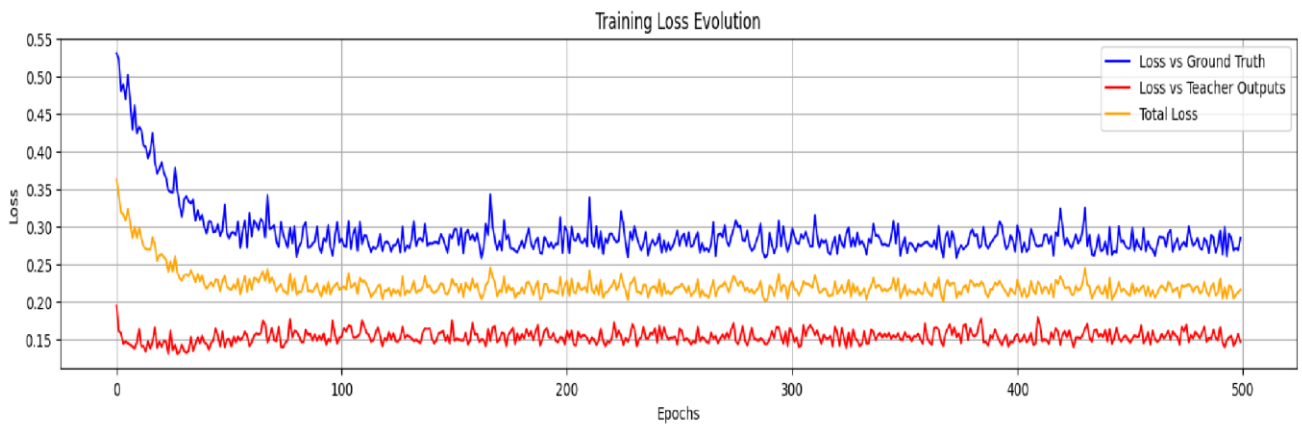
In a second step, I used this same model as a student by associating it with a teacher provided by another researcher in the laboratory. This teacher model had been previously trained, but its performance was relatively poor, with metrics well below what is expected for a reliable emotional recognition task. Knowing this, I expected that distillation from this teacher would not bring any significant improvement, or even that it could degrade the student's performance. I nevertheless conducted the experiment in order to validate this hypothesis and observe the impact of a poor quality teacher on the student's learning. This step is part of an exploratory approach, aimed at better understanding the limits of distillation in a cross-modality context. Note that the metrics shown below are only for the teacher model, not the teacher–student combination. They serve to illustrate the initial weakness of the source model used for distillation.

Metrics	MFA	MSE	R2_Score	CCC
Valencia	0.6574	1.4094	-2.1470	-0.0202
Arousal	0.6257	1.2110	-1.5911	0.1528

In our knowledge distillation setup, we observed an important problem: the predictions of the teacher model are not aligned with the labels and in some cases, they have no relation to reality.

In summary, the teacher model behaves almost randomly, learns nothing meaningful, and its influence could potentially harm the student's learning if he or she depended on it too much.

Even though the loss function with soft labels may seem lower, it doesn't mean that these predictions are close to reality. They are simply easier to imitate.



Metrics	MFA	CCC
Valencia (Val GT)	0.6454	-0.0771
Arousal (Val GT)	0.7103	-0.0285
Valencia (Val Teacher)	0.3436	0.2567
Arousal (Val Teacher)	0.3920	-0.0641
Valencia (GT Train)	0.3988	0.5975
Arousal (GT Train)	0.4779	0.5630
Valencia (Train Teacher)	0.3268	0.3992
Arousal (Train Teacher)	0.2870	0.5163

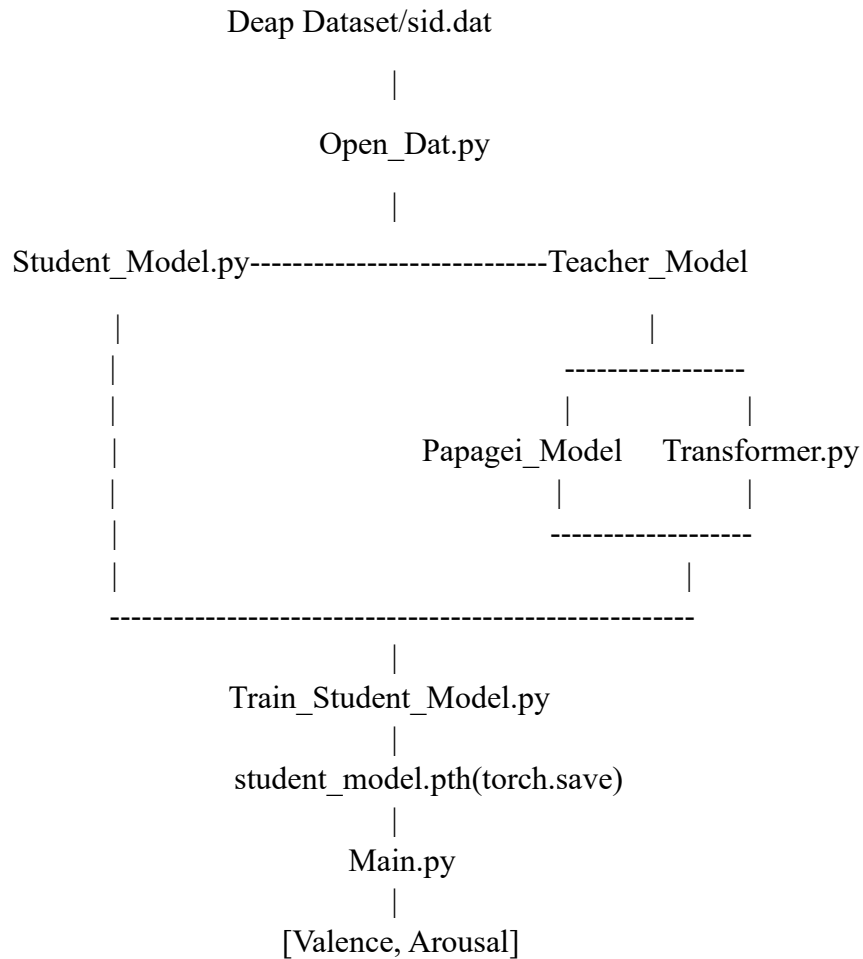
The results show that the model continues to perform well on training data, with moderate MAEs and CCC scores around 0.5 to 0.6, especially when compared to true values (GTs). On the other hand, on the validation data, the errors are significantly higher and the CCC scores are close to zero or even negative, indicating a low correlation between predictions and true emotions. Even the predictions in relation to the teacher's outputs remain limited, with a low CCC. This confirms that the model has learned to align with training data, but struggles to generalize. The analysis of the graphs allows us to visualize this overfitting trend.

5.3.2.3.3. Student_Model_v2/3

Following these observations, I decided to work more with the student model alone in order to test two points. First, I tried to check if it was possible to make predictions in real time via the Apple Watch. It worked technically, but due to the low accuracy of the model and the fact that the test data came only from myself, I couldn't confirm if the predictions were actually reliable. Then, I tested an approach where the model was only trained on one emotional dimension at a time (e.g., only valence), instead of learning valence and arousal simultaneously. The goal was to see if it allowed for better alignment with the labels. However, this dissociation also did not significantly improve performance.

5.3.3. DEAP Dataset

5.3.3.1. Code Architecture



5.3.3.2. Model information

Epochs	3000
Batch Size	64
Learning rate	0,001
Weight Decay	1st-3
Patience	50
Patience Scheduler	10

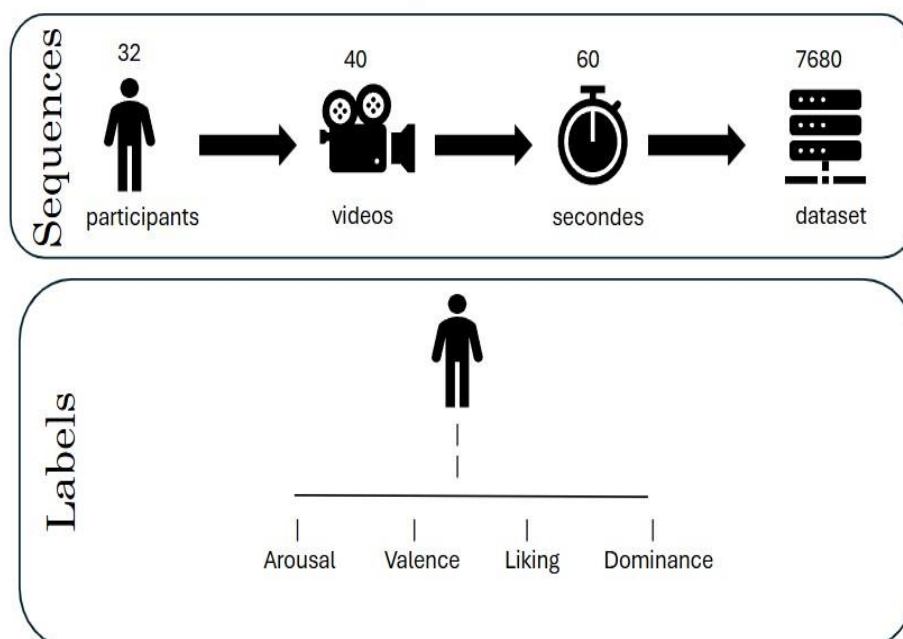
Shuffle	True
Dataset size	7680

5.3.3.3. Results

5.3.3.3.1. Teacher model

In the rest of my project, I initially used the YAAD dataset, an ECG dataset, as well as a *teacher* model developed by a collaborator in my laboratory, also trained on this dataset. However, the final objective being to detect the emotional state from the heart rate extracted from Apple Watches, a problem arose: the method of measuring the heart rate differs between the ECG (electrocardiogram) and the PPG (photoplethysmography) used by the Apple Watch. Even if the differences are minimal, they affect the way heart rate is calculated, making training the model on ECG data irrelevant for an application based on PPG data. For this reason, I decided to change my approach and rely on a model trained on the DEAP dataset, which contains physiological recordings including PPG signals, and therefore more consistent with our objective. This change of dataset involved a complete revision of the *teacher* model used for knowledge distillation. I then chose to use Papagei, a pre-trained open-source model for multimodal video analysis, as a new *teacher model*.

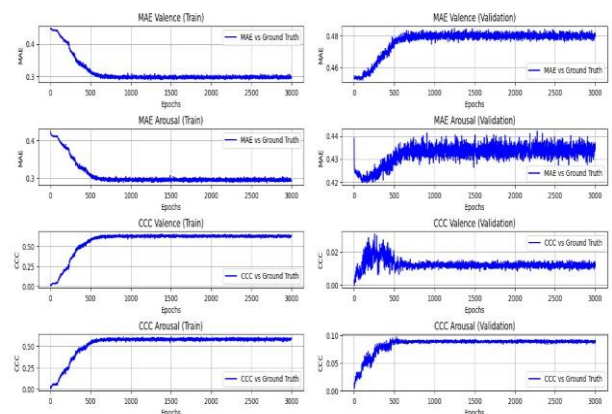
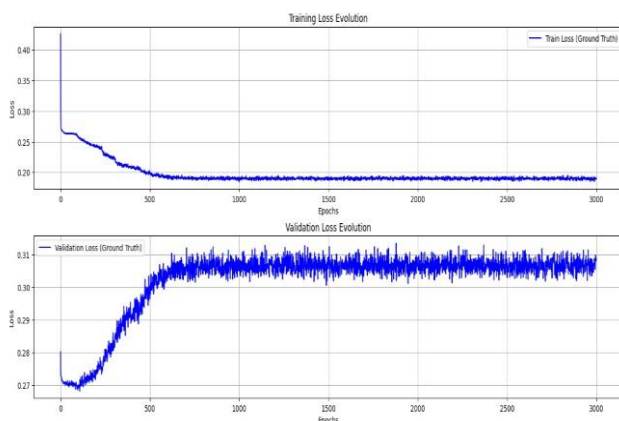
Deap Dataset



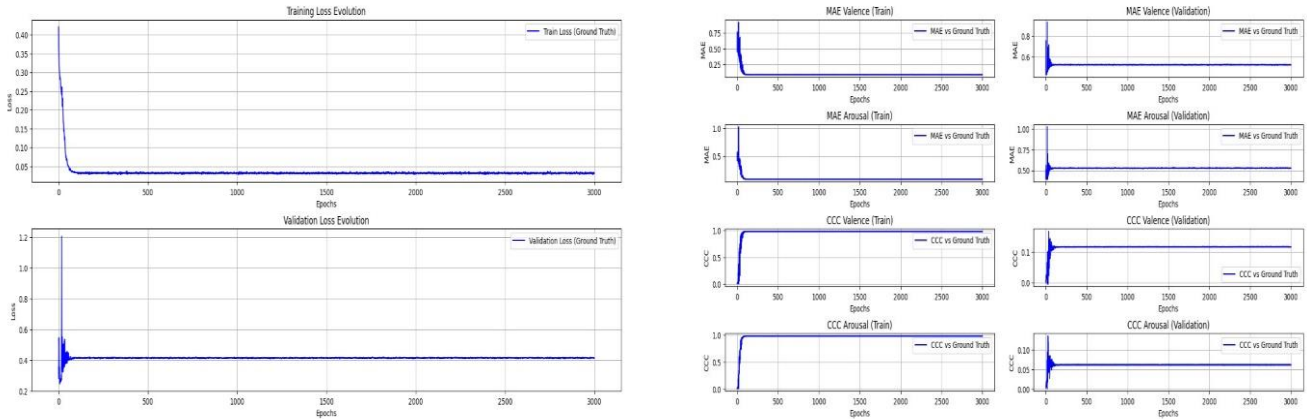
The DEAP dataset contains recordings of 32 participants who each watched 40 60-second videos, for a total of 1280 videos. Each video is annotated with four emotional dimensions: *valence* (level of pleasure felt), *arousal* (level of awakening or activation), *dominance* (feeling of control over the situation) and *liking* (overall appreciation). In our case, only the valence and arousal scores are used for the prediction. In order to increase the amount of data, each video has been divided into 6 segments of 10 seconds, each of which inherits the same labels as the full video, giving us a final dataset of 7680 segments. The Papagei model provides embeddings, i.e. dense vector representations summarizing the multimodal content of each segment. These embeddings are then used as inputs for a small model responsible for predicting valence and arousal scores, as Papagei is not directly designed to produce these output scores.

For this prediction, I tested two architectures: an MLP (Multi-Layer Perceptron) and a lightweight transformer. MLP is simple and effective for dealing with dense vectors like embeddings, but it does not take into account the internal relationships between the different dimensions of the vector. Transforming it, on the other hand, uses a self-attention mechanism, which allows the model to weight the relative importance of each component of the embedding relative to the others. This gives it a better ability to capture complex interactions in data. In my experiments, the transformer did indeed perform better than the MLP, justifying its use despite its slightly higher complexity.

MLP



Transform



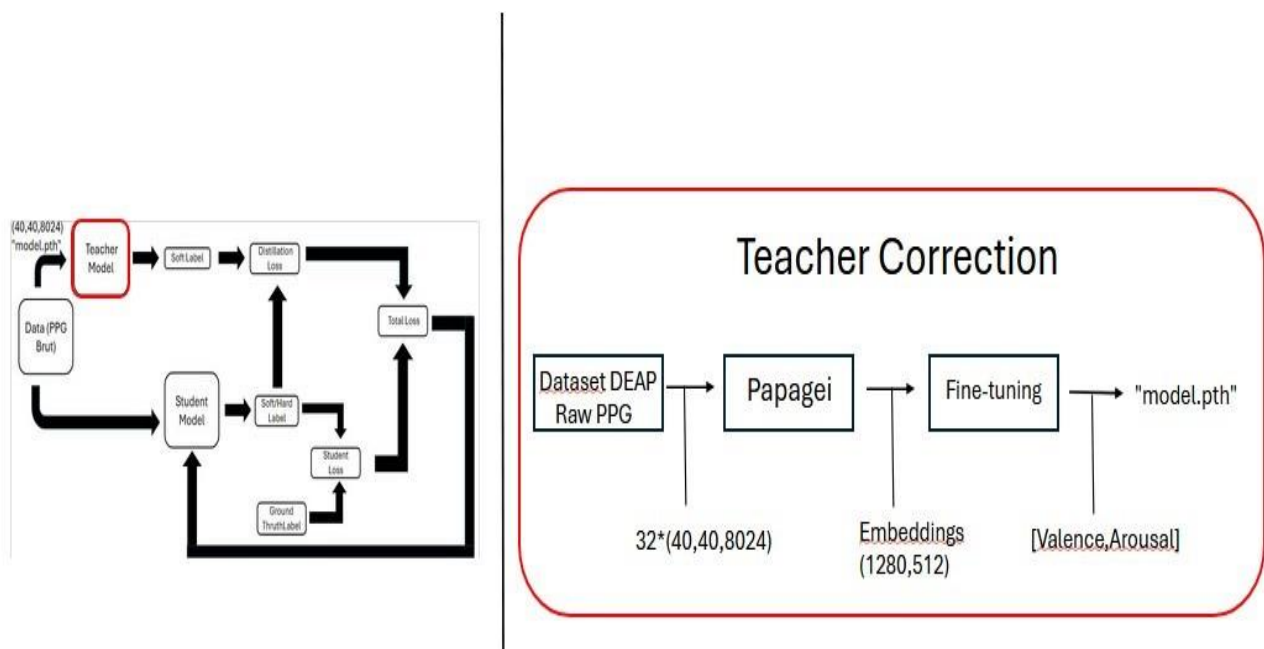
MLP Metrics	MFA	CCC
Valencia (Train)	0.3186	0.5574
Arousal (Train)	0.3130	0.5102
Valencia (Val)	0.4655	0.0215
Arousal (Val)	0.4272	0.0777
Transformers Metrics	MFA	CCC
Valencia (Train)	0.0789	0.9795
Arousal (Train)	0.0832	0.9753
Valencia (Val)	0.5252	0.1164
Arousal (Val)	0.5275	0.0612

The MLP manages to capture a certain level of correlation across the training set, with CCCs around 0.5. However, its performance drops sharply in validation, especially on the CCC (< 0.1), which suggests overfitting and poor generalization. The Transformer shows exceptional performance in training, with CCCs very close to 1, which indicates excellent memorization capacity. However, just like the MLP, its

validation results are much lower, especially on the CCC, which also reveals a strong overfitting. Both models learn training very well, but fail to generalize to new data.

The results I have just presented correspond only to the performance of the teacher model, i.e. the Papagei model associated with the small predictive model (MLP or transformer) used to estimate valence and arousal from embeddings (Fine-Tuning).

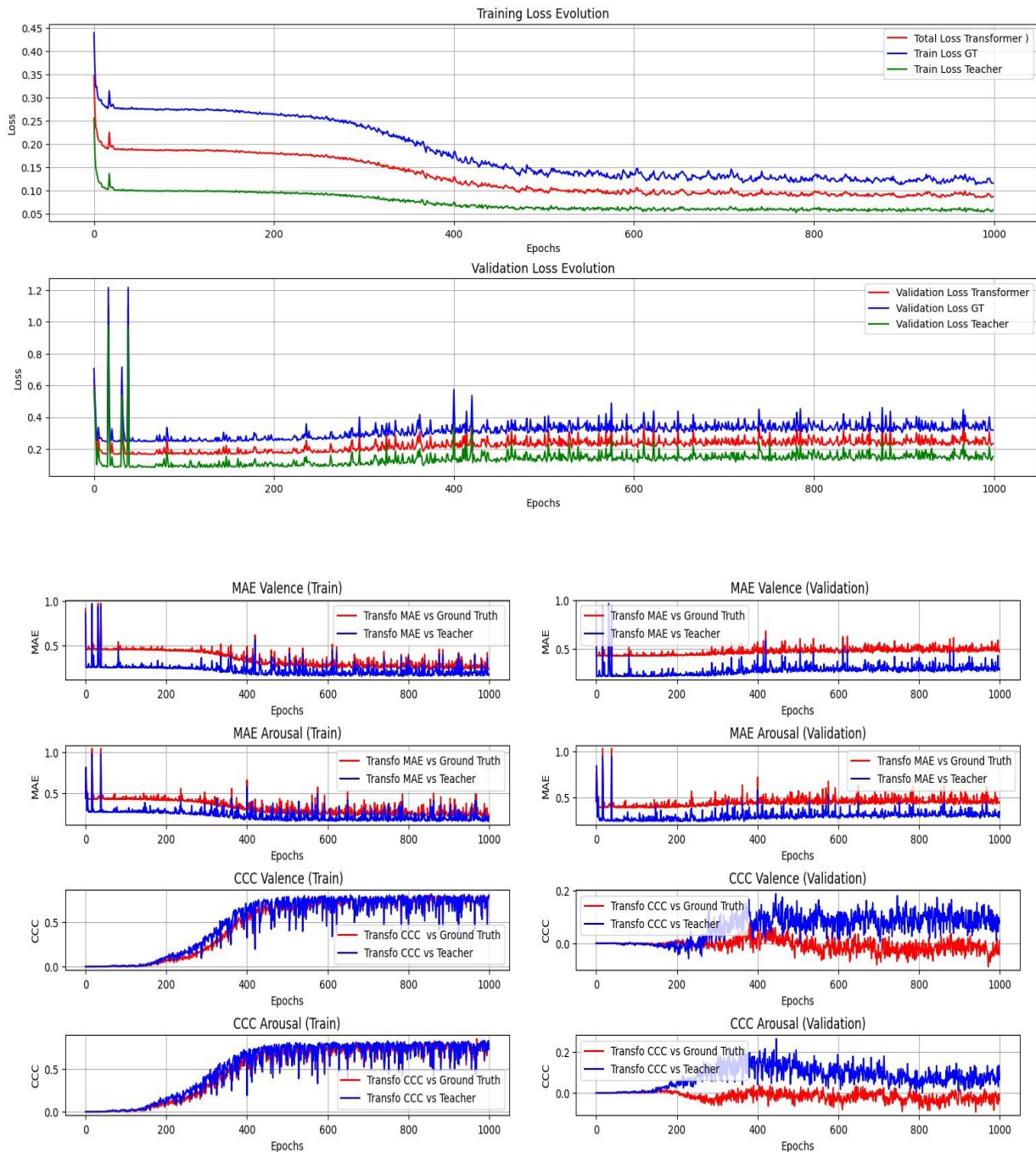
The knowledge distillation phase, where this knowledge will be transferred to a lighter model (*student*), will take place in the following stages of the project.



Finally, I realized that another lab member had already worked on a similar part of the project. Because of a miscommunication, I was not aware of it and I followed my own approach. Fortunately, our results turned out to be very similar, which gave me confidence in the validity of my pipeline, which I therefore chose to keep.

5.3.3.3.2. Teacher + Student model

So I chose to use a transformer both for fine-tuning and as a student model in the context of knowledge distillation, with Papagei as my teacher model. The results obtained for these different configurations are presented below.



Metric (Val)	MAE T	MAE GT	CCC T	CCC GT
Valencia	0.3405	0.5214	0.0389	0.0192
Arousal	0.3153	0.4604	0.0980	0.0588

For valence, the model obtains an MAE of 0.3405 when the predictions are compared to the teacher's outputs (T) and 0.5214 when compared to the reference values (ground truth, GT). The concordant correlation coefficients (CCCs) are 0.0389 (T) and 0.0192 (GT), respectively.

For arousal, the MAE is 0.3153 (T) and 0.4604 (GT), while the CCCs reach 0.0980 (T) and 0.0588 (GT).

These results indicate that the model is closer to the teacher's predictions than to the actual labels, with lower errors and a slightly higher CCC in comparisons with T. In the end, these performances remain not very encouraging and suggest that improvements are needed for cross-distillation to bring a real benefit.

6. Astonishing Report

During my internship, I had the opportunity to discover a professional and scientific environment that was very different from what I had imagined, especially in comparison to my academic training.

A first surprise was the great freedom in the organisation of working time. Unlike a very structured and structured schedule in class, I was able to set my own schedule, which required a great deal of autonomy and personal discipline. This flexibility is a significant advantage, but it also requires good management to remain effective.

I have also noticed that the working method of researchers differs significantly from that taught in engineering training. Where teaching is often very guided, with well-defined objectives and steps, research involves much more exploration, testing, and sometimes trial and error. This less linear approach forced me to adopt a certain open-mindedness and patience in the face of sometimes uncertain results.

Finally, I was pleasantly surprised by the ease and fluidity of the exchanges within the laboratory. Informal discussions with colleagues, the availability of researchers and the collaborative atmosphere greatly promote understanding and the advancement of projects. This contrasts with some of the more formal settings I may have known in class or other environments.

These differences have allowed me to become aware of the human and organizational skills that are essential in a research environment, beyond technical knowledge. This internship was therefore very rewarding, both professionally and personally, and offered me valuable insight into how a laboratory really works.

7. Conclusion

On a technical level, although I was not able to complete all the work required during this internship, including the development of a fully operational AI capable of detecting an individual's emotional state on a mobile device, the results obtained nevertheless provide a solid basis for the continuation of the project. As explained during the various weekly presentations, the progress made aroused a certain interest among the researchers, who underlined the relevance and quality of the work accomplished. During the last presentation, a professor from ENSEA, Mr. Alexandre Pitti, even came to congratulate us, which was a real recognition of our involvement and the results obtained.

On a personal level, this experience allowed me to confront the reality of a research project, with its constraints and unforeseen events, while benefiting from the support and encouragement of a passionate team. In addition, I was offered to continue this collaboration by contributing to the writing of a scientific publication, which represents a great opportunity to deepen my skills and promote this work in an academic setting. This internship was therefore very rewarding, both in terms of the acquisition of technical know-how and the discovery of the research community.



